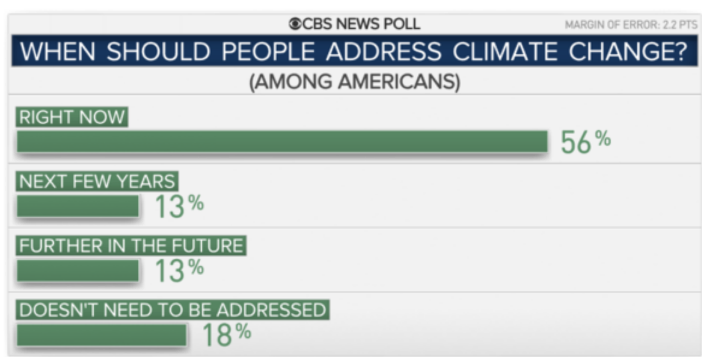
None of these issues can be solved by a single entity, and neither can climate change.

A TALE OF THREE CITIES

It is rare that we are presented with a problem that affects everyone. This past year we unfortunately have become quite familiar with complex, all-encompassing problems. The spread of the coronavirus has shed a light on rampant injustice, wealth gaps, and racial inequalities, just to name a few. In the hierarchy of all-encompassing issues, it can be argued that climate change is at the top of the list. Its arms extend to every corner of the earth, and its fingers will disrupt every industry and way of life to which we have become accustomed in the past century. Although climate change can and will affect all of us if we give it the chance, many feel as though it is too big a problem for them to care about solving, or one that we simply cannot solve. According to a CBS poll, only 56% of Americans believe that climate change should be addressed now, with 26% saying it should be done in the future, and 18% flat out denying that it needs to be addressed at all. The article this graph is from has an optimistic tone, saying that we have reached an important milestone now that a majority of Americans believe we need to take action. I will argue that a simple majority is not a triumph, because, as is a key part of this paper, climate change can only be curbed if we have 100% of the earth's population working together. The correct response to the question "when should people address climate change" does not appear on the graph above. The correct response is, "*why have we not started addressing it already?*"



In Europe and other parts of the world, we have seen a push towards a sustainable future coming from the highest levels of government. The United States federal government in comparison has fallen behind in recent years, with state governments by and large following suit. In the absence of leadership at a federal and state level, we have seen individual cities step up to the plate. The three cities being examined, Philadelphia, New Orleans, and Phoenix, each have a very different relationship with climate change, and very different approaches to doing their part to address it.

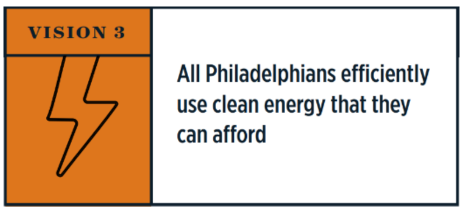
The first city we will be examining is Philadelphia, PA. The city aims to be a beacon of sustainability in the region if not the country by being one of the first American cities to create a comprehensive plan outlining a future where not only is the planet is healthier, but people are healthier too. This planet-plus-people method has been adopted and adapted throughout the country.

Compared to Phoenix and New Orleans, Philadelphia's relationship with climate change is a rather tame one. It is not prone to hurricanes, forest fires, or any other extreme natural events, for that matter. It is also not coastal, and so it will take longer for it to feel the effects of a rising ocean. Really, the only thing Philadelphia has to lose in the next two or so decades is snow, as climate change will make its winters warmer and wetter. Even though this is the case, the people who live in Philadelphia support climate initiatives by and large, with 80% of Philadelphians believing that climate change is happening right now, according to a map produced by Yale in 2020. This is 8% higher than the national average.

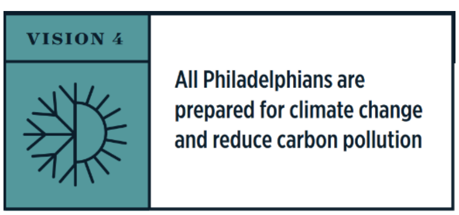
In Philadelphia, the plan to become sustainable, called Greenworks, "acknowledges and addresses the local impacts of the biggest environmental challenge of our generation: climate change" (Greenworks, 3). Established in 2016, the plan is not the first time Philly has openly addressed climate change and made ambitious plans to combat it. In 2007 during his inaugural address, now former mayor of Philadelphia Michael Nutter stated that during his tenure as mayor he would, among other things, focus on making Philadelphia into the "greenest city in America." This was a tall order, considering Philly was and still is not the easiest city to convert to a greener future. The city has old infrastructure, inefficient buildings, a dying public transit system, and some of the highest poverty numbers in America. One thing that Philadelphia does have, though, is a wide support of green initiatives in their city, and as Greenworks is quick to point out, "planning for the changes we know are coming must be a priority for Philadelphia's government, businesses and institutions, *and* residents" (Greenworks, 3). Something that has garnered a lot of support for Greenworks is its acknowledgment that a greener Philadelphia only works if every neighborhood and zip code feels the positive effects of the changes taking place, so they created four categories than not only aim to improve the health of the world, but do so while also improving the health of its citizens. These four categories are Equity, Health, Environmental, and Economic. In order to show improvement in all of these categories the plan cites eight areas called "visions" where improving the sustainability will directly improve people's lives.

In a truly sustainable future, a city's watershed is cleaner, which leads to cleaner drinking water as well. According to Greenworks, "In 2014 22% of Philadelphians were food insecure," and so, while making the infrastructure of Philly greener, we also have to make the grocery store shelves greener.

Air quality is a subject area that we are used to hearing about these days, especially in places like California. But what people neglect to realize is that the air quality in Philadelphia is not good and it's getting worse. "In 2015 Philadelphia had 16 unhealthy Air Quality Index (AQI) days" (Greenworks, 4). Cleaning up the energy production and transportation practices of the city will help fix this problem by default for the citizens.



Clean energy is more affordable than nonrenewable energy. An article published in Forbes in 2019 states that, "According to a new report by the International Renewable Energy Agency (IRENA), unsubsidized renewable energy is now most frequently the cheapest source of energy generation" (Greenworks, 5). Cheaper energy production and less expensive maintenance means cheaper energy bills for the consumer.



The authors of Greenworks know that even if Philadelphia reaches its most lofty environmental goals, climate change is a worldwide issue that needs to be combated by every city on Earth. "In 2014, 53,138 sensitive Philadelphians lived in hotter than average neighborhoods" (Greenworks, 5). In the event that climate change is not counteracted on an international level, Greenworks hopes to prepare the citizens of Philadelphia for a post climate change world.

Philadelphia is home to some of the most impressive and beautiful urban parks in the country, but many of its smaller neighborhood parks need greater care and attention. A greener city where trees suck up CO_2, storm water is managed and filtered, and the waterways are healthy is a strong city. The city started work on this vision by installing green stormwater infrastructure that catch rain falling on 838 acres as of June of 2016.

Low carbon transportation is a must in a modern city, especially if it is the sixth largest city in America. Cars are still dominant in the city, but in recent years more people have chosen to bike to work or ride public transit as the city invests more money in updating SEPTA. While rail travel has been increasing, bus travel has seen a decline in the last 7 years in almost every neighborhood across the city. Also, “About 4,000 more survey respondents said they commuted by bike than reported traveling that way in 2010 (Philadelphia Inquire). Even with these improvements, in 2014 only 37% of Philadelphians chose a low-carbon commute (Greenworks 7). Public Transit is an area in which Philadelphia has the opportunity to excel due to the fact that SEPTA is actually a rather robust transit system of which many older cities are jealous. Even though many of the older narrower streets make it more difficult to add things like bike lanes, not all of the cards are stacked against Philly.

When I asked former Mayor Nutter what the largest hurdle is that Philadelphia has to overcome in order to become a green city, he said waste management. According to Greenworks, Philadelphians generated 2.5 million tons of waste in 2014. Recycling in America in general is poor and not at all at the levels it needs to be. That is an issue that even a city as large as Philadelphia cannot solve on its own. The state and country need to become more invested in our recycling infrastructure. That being said, on a city level legislation has been passed putting harsher restrictions on littering and hiring more personnel to keep the streets clean. Education about recycling is also becoming more of a priority. The goal of Greenworks is to eliminate all waste in Philadelphia, and, in order to do that, they "must significantly reduce the amount of trash we create and increase the amount we reuse and recycle." Easier said than done.

Even plans far less ambitious than Greenworks require participation from everyone in the city. That means Philly will need to educate the public on why and how to get involved, convince businesses to become sustainable, and, while doing so, create jobs in the area of sustainability. The city has seen success in this area, with residents from 92% of Philadelphia's zip codes participating in the updating of Greenworks. Some examples of newly created jobs in the environmental sector, according to the website Buskowitz, are:

Sustainability Consultant	Sustainability Director	Environmental Engineer
Pollution Control Officer	Water Engineer and Scientist	Green Building Professional
Agriculture and Food Scientist	Renewable Energy Engineer	Smart Grid Engineer

From the information above we can gather that Philadelphia is attacking the subject of climate change from all angles, in hopes that by improving the climate crisis they can also improve the aspects of life in the city in general, and set it up for a healthy, sustainable, and prosperous future. I think that relating sustainability to the average resident of Philadelphia is a good strategy because in a city that is crippled by so much poverty, hunger, substance abuse and other major issues, the community will only get on board if they can see how building a greener city doesn't just save the environment, but could also vastly improve their quality of life.

The steps that Philadelphia is taking are big, and, if successful, will set the gold standard for other American cities in the future. But they have fairly little to lose in a climate change world compared to the other two cities that I will be analyzing.

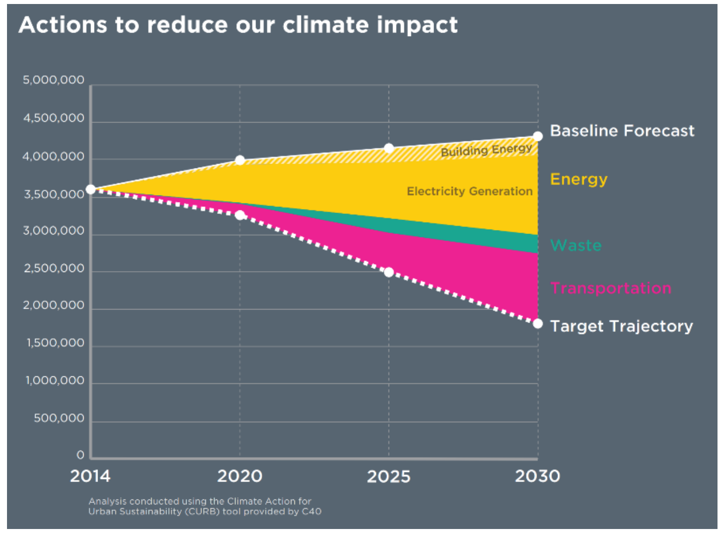
While Philadelphia is trying to prevent a world where summers are hotter and winters are wetter, New Orleans is trying to prevent a world where the city ceases to exist. In New Orleans, the Greenworks equivalent is the "Climate Action for a Resilient New Orleans." In this extremely vast and all-encompassing document, the city of New Orleans lays out its plans to reduce its carbon emissions by 50% by 2030. The document is based on two vision statements, and they are as follows:

- In 2030, New Orleans will have reduced our annual greenhouse gas pollution by 50% from what it is today. We will use 100% low-carbon electricity, take 50% of our trips in modes other than driving, and divert 50% of our waste from landfills.

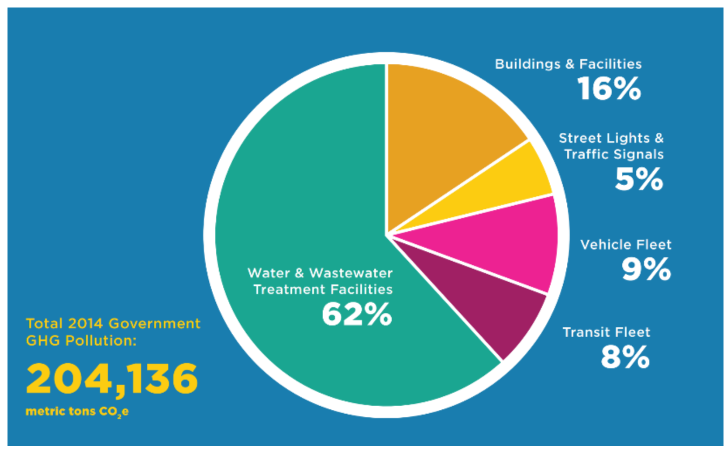
- We are a dynamic, prepared, and equitable city embracing our changing environment and working together to combat climate change and slow its effects. Our economy and quality of life benefit from this investment in curbing climate change and being more resilient.

According to the plan, “In August 2015, New Orleans became the first city in the world to release a comprehensive resilience strategy outlining actions to address environmental, social, economic, and infrastructural challenges.” The document is very straightforward and has a different, more dire tone than that of Greenworks. From the very beginning, it states that even with every preventative action, there is no future where New Orleans does not lose any coastal lands and properties. Similar to Greenworks, a large portion of the Climate Action plan is dedicated to trying to sway public opinion in favor of sustainability. The New Orleans plan does this partly by using fear as they explain what is at stake for the city, but they also spend large amounts of time talking about using this movement as a way to alleviate poverty and make New Orleans more resilient. “Taking action on climate can help us address some of those stresses with the opportunity to alleviate poverty and unemployment while improving neighborhoods and community connections” (Climate Action, 17).

Something unique about this plan is how often it states how little we know about what the future will bring. Resilience is the most important word in the document, and they define it on almost every page. Resilience in New Orleans means being able to adapt and accepting the fact that the future is unpredictable and the city will have to make broad, sweeping changes in short periods of time in order to combat it. Until page 20, there is not a single mention of what will be done to accomplish their goals.



The plan utilized many charts and visually stimulating media in order to show the projected goals of the city. Greenworks relied more on writing than the Climate Action plan does.



These are the categories that New Orleans has chosen to focus on:

- Modernizing Energy Production and Use
- Improve Transportation Choices
- Reduce Waste
- Creating a Culture of Awareness and Action

To modernize energy production, the city is not doing anything out of the ordinary. They are shutting coal powerplants down completely, building wind and solar farms, and updating existing buildings in order to make them more efficient. Right now, they receive 4% of their energy from coal. In order to get this number to 0%, the city is working closely with the leading producer of coal fired energy in the region, Entergy. Entergy has committed to slowly transition to renewable sources of energy in an attempt to change with the times. They have taken it upon themselves to close one of the powerplants that supplies energy to New Orleans by 2028, and the city council of New Orleans is very happy to create green initiatives by then to offset that energy gap. One interesting subject area that is covered briefly in this section is how much energy New Orleans expels controlling the water in and around the city. The city has an entire power plant devoted to powering these pumps, some of which can move water at a higher flow rate than the Ohio river. Even with all that energy generation, it still requires more power from additional power plants, especially when the thirteen under-highway pump stations are automatically turned on when it rains.

By 2030, they want 50% of transportation within the city to be emission free. To accomplish this goal, they intend to increase the bike lane network, and over the past eight years they have added over 100 miles. Streets are also being redesigned in order to make them more walkable, and more desirable. After hurricane Katrina, much of the bus fleet in New Orleans was replaced, but not a single electric bus was purchased due to the stress of the times. Every 12 years, the city looks to purchase new buses, and so this year in 2020 they are on track to buy many electric buses when they update their fleet.

The waste management portion of the document is, in my opinion, the most ambitious. By 2030 they want to reduce the landfill waste by 50% and bring that number to 100% by 2050. The city plans to build facilities that turn the methane produced by the decomposition of the waste that cannot be recycled into usable clean natural gas. They say that the investment in technologies like this will create thousands of jobs while keeping emissions out of the air. Presently, only 41% of residence as of 2017 had requested recycling bins and received them from the city. This number needs to increase and New Orleans thinks they can make than number 100% through the use of education.

A key part of education is cultural awareness. New Orleans plans to connect culture with climate action while growing the local low-carbon economy. They plan to utilize a data driven initiative to make collaborative decisions throughout the city. The vison statement for this section is to, "increase awareness and action among residents, businesses, and visitors."

I appreciate how they mention visitors in this goal because anyone who knows New Orleans knows that tourism is a huge player in the city's greater economy with 17.7 million visitors who spent $8.7 billion in the city in 2017 alone (neworleans.com). The architects of the city's green culture believe that one of the best ways to increase cultural sentiment in favor of sustainability is to create thousands of jobs in the green sector. "According to the 2017 U.S. Energy and Employment Report from the Department of Energy, solar jobs represent more than 50% of the electricity generation jobs in Louisiana. A December 2016 report from E2 estimates about 25,000 Louisianans are employed in energy efficiency jobs, the majority in heating, ventilation, and air-conditioning. A 2015 solar jobs census from the Solar Foundation noted the U.S. solar installation sector employed 77% more people than the domestic coal mining industry, and created more jobs than oil and gas pipeline construction and crude petroleum and natural gas extraction combined in 2015. The 2016 update to the solar report noted that one out of every 50 new jobs added in the United States in 2016 was created by the solar industry, representing 2% percent of all new jobs and that Louisiana has 2,922 solar jobs statewide, ranking 24th nationally and 18th per capita" (Climate Action 61).

Although long, that quote is a testament to the work that the city has put into creating jobs in the sustainability sector.

Creating jobs isn't the only way New Orleans is getting the community involved in green initiatives. One example is hosting tree planting benefits. They have utilized over 85,000 volunteers to replant 60,000 of the 100,000 trees lost during hurricane Katrina with the goal of a net increase of 40,000 trees by 2030.

New Orleans is sinking in water, but the last city that we will be evaluating has been drying up even before climate change began. The "Climate Action Plan for Government Operations" is the main plan created by the government of Phoenix, and, compared to Greenworks and the Climate Action Plan of New Orleans, it is a little sparce. Weighing in at 27 pages compared to the 76 in Climate Action, the most recent update of the document was written in 2009. Many of the goals presented seem to be quite tame, and only enforce change within the municipalities of the Phoenix government, made clear by their goal statement: "Reduce emissions from city operations to 5 percent below the 2005 levels by 2015."

The situation in Phoenix really is more dire than people tend to realize. The state government of Arizona found in a report than western North America's average temperature has risen an average of 1.4 degrees Fahrenheit over the past 50 years. The higher temperatures have been directly linked to lower water levels in reservoirs, lakes and streams, as a result of increased evaporation. There has been a 15% decrease in runoff, a 40% decrease in water basin storage, and a projected shrinkage of 15-30% of Arizona's forest land, not to mention farmland, which is much less resilient to drought. The government of Phoenix describes its role in curbing climate change by stating that they can "reduce the emissions from city operations by increasing energy efficiency in buildings and vehicles, expand alternative fuels, utilize renewable energy sources, waste reduction, and promote alternative modes of transportation for employees" (Climate Action Plan 8).

The city is attempting to create more public transit opportunities with the construction of a 20 mile electric rail service, improving building codes to require more stringent environmental codes, changing the zoning of the city to make it more pedestrian friendly, increasing the amount of "urban forestry", and using water chilled overnight to cool buildings during the day. All of these initiatives are exciting, but what they lack are the hard numbers and goals that the other two reports we looked at have. One interesting area that Phoenix spends some time taking about is traffic systems. Traffic systems and streetlights account for 7% of the energy usage in the city. The report explains that replacement of regular traffic lights per intersection costs anywhere between $10,000 to $11,300, factoring in labor and other costs. While other cities are hoping to get their emissions cut in half in the next ten years, Phoenix was hoping to change 40% of their traffic lights to LEDs by 2005, lowering the amount of energy these lights use by 54%. These do not seem like equal contributions to the fight against climate change.

They are using a mix of 20% biodiesel in their non-transit vehicle fleets. Biodesal does still release greenhouse gases, but they are quick to point out that, since the plants it is made from absorb carbon while growing, no more carbon is released into the air. While this may be true at the burning of the fuel, they do not mention the CO_2 released during the farming, processing, and refining of biofuel. According to Greentechmedia.com, "creating biofuels like corn ethanol is an energy-intensive process. The steps involved with growing a feedstock like corn on an agro-industrial scale -- think farm equipment, fertilizer, harvesting, transporting the feedstock to the biorefinery, converting the feedstock into ethanol/biodiesel, and further transporting the biofuel to a petroleum refinery or service station -- require a tremendous amount of energy, especially when more than 10 billion gallons of biofuel are being produced in any given year in the United States." This is not to mention the hundreds of thousands of acres of forests being cleared in order to produce biofuel crops. With a growing population, we need to use all our limited farmland for food that we eat, not food that powers our "non-transit vehicle fleets.".

Phoenix hopes to trap much of the methane that it produces in its landfills and use it as a clean source of energy. The bar they have set for themselves is to "enhance methane collection systems at two landfills: Skunk Creek and SR-85 … [in order to achieve] incremental improvement for the methane collection efficiency which goes beyond the industry standard." (Climate Action Plan 24). "Beyond the industry standard" seems like a goal that is too low for a city in such a dire situation. The Washington Post reports that if things continue as they are, Phoenix will by uninhabitable by 2050. The Guardian is quoted as calling Phoenix "the world's least sustainable city" and "and urban bullseye for global warming." With snow melt in the Rocky Mountains being 70% lower than in previous years, Phoenix has almost no more cards to play in its fight.

It would be unfair to say that the people and government of Phoenix do not care about the environment at all, but it is not just their lack of climate action that speaks volumes, it is their current anti climate actions that are most concerning. Outside of Phoenix Arizona there are several developments that have sprung up in recent years, like Anthems. Built in the desert, nothing is natural about these developments, including their large lakes, resort pools, and golf courses. Looking at these walled communities, it would be hard to imagine that this area of the United States is short on water, because it is being used with reckless abandon. With its lush vegetation, you would never know that Phoenix gets only eight inches of rainfall each year. It gets its water form two lakes, Lake Mead, which is 300 miles away, and a closer lake, Pleasant, which resides on a Native American Reserve. Both of these lakes are fed by the Colorado river, which is trending smaller and smaller every year. Phoenix “is a modern shrine to towering concrete, and gives way to endless sprawl that stretches up to 35 miles away to places like Anthem. The area is still growing – and is dangerously overstretched, experts warn” (TheGaurdian).

Continuing on its upward trend, there are plans for substantial growth in the coming years and there just isn’t the water for it, according to Jonathan Overpeck who authored a 2017 report that linked declining flows of the Colorado River to climate change. These plans include Bill Gates’ “smart city” which will sport 80,000 new homes west of Phoenix, and a “masterplanned community” modeled after the hilltop town of Tuscany. It will have five golf courses, a vineyard, parks, lakes, and 28,000 new homes. “The promotional video contains no details about where all the water will come from, but boasts: “This is the American dream: whatever you want you can have” (TheGaurdian). Even with the Federal Bureau of Reclamation reporting in 2012 that droughts of five or more years would begin to happen every decade over the next fifty years, Phoenix and the surrounding areas have put into effect no water restrictions.

The energy usage in Phoenix is astounding. Most of the energy is used by the water department which uses electricity to pump water uphill from the Colorado to Phoenix and Tucson.

Phoenix itself has become rather good at recycling its water, but almost all of its recycled water goes directly to cool the largest nuclear powerplant in the United States, where it is then discarded. In addition to being the largest nuclear powerplant in the US, it is also the only one not located on its own body of water, and, for this reason, all of the water for its cooling must be brought to it at great expense. Until 2019, most of Phoenix's energy still came from the large Navajo Generating Station which is a coalfired powerplant. This plant was close in 2019, but was not replaced by renewable sources. Even though, in 2018, it received more than 330 days of bright sunshine every year, Phoenix only receives 2-5% of its electricity from solar power.

"In his 2011 book, Bird on Fire, the New York University sociologist Andrew Ross branded Phoenix the least sustainable city in the world. He says he stands by his assessment and warns of an "eco-apartheid," whereby low-income neighborhoods on the more polluted south side of the Salt River (which once flowed vigorously through the city and is now a trickle) are less able to protect themselves from the heat and drought than wealthier citizens. "There's a stark disparity," he says. "The resource havens, with their hybrid cars, their solar panels and other green gizmos; and the folks on the other side struggling to breathe clean air and drink uncontaminated water. It's a prediction of where the world is headed." But we do not have to look into the future to see where Phoenix and other areas like it are heading. The Hohokam peoples were the original settlers if the region where Phoenix now resides, and their civilization with just 40,000 people collapsed in the 15th century over disagreements on how to divide the scarce water supply.

According to Jim Holway who is the vice president of the Central Arizona Water Conservation District, "climate change is having an impact but that's a controversial, unsettled issue in the western US".

There are many excuses that could be made for this city's relative lack of climate initiative: the underfunded schools in poor Black and Latino communities, the already stressed city budget, a population that is growing faster than the city can keep up, etc. But every city is dealing with issues of their own, and in a climate plan that complains about the cost of things as simple as LED traffic lights, it is hard to understand why Americas fifth largest city is so behind in the fight for a sustainable future when they have the most to lose.

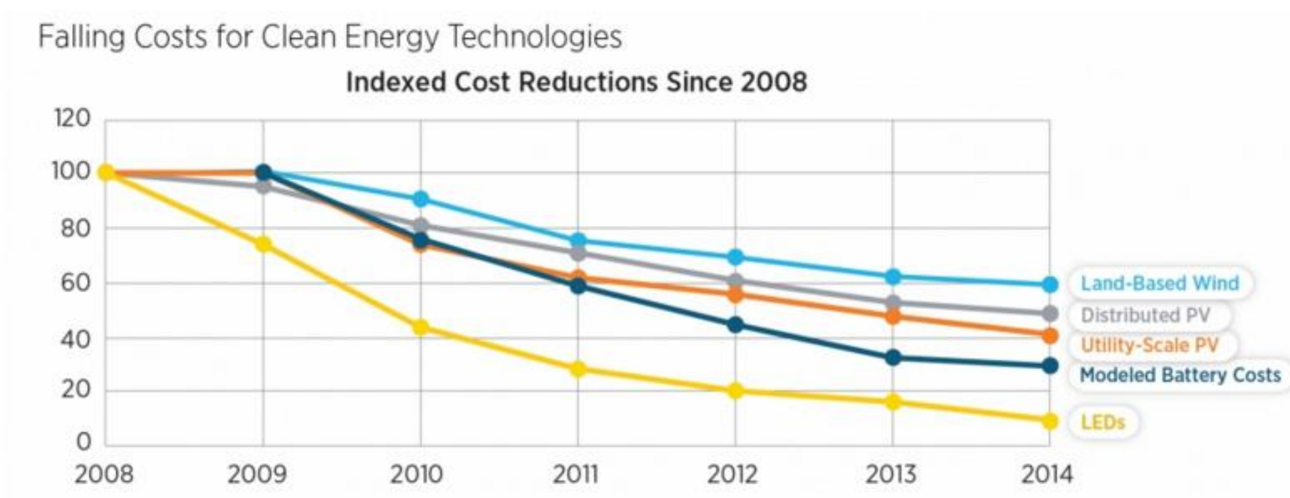
We now see that some cities are more ambitious than others in their climate initiatives, and this ambition does not seem to correlate with how dire the consequences of their inactions are. Just in the case studies examined in this paper, we see a large disparity between the attention to climate in the plans for Philadelphia, a city in a good geographic position for a post climate change world, and Phoenix, which is the opposite. Many have tried to answer the question of why this is. It would seem like water restrictions would be a commonsense initiative for places like Phoenix, yet we have yet to see a single one. This may have something to do with the psychological idea of willful blindness. Margaret Heffernan is a leading researcher on this subject and has examined the "intricate, pervasive cognitive and emotional mechanisms by which we choose, sometimes consciously but mostly not, to remain unseeing in situations where 'we could know, and should know, but don't know because it makes us feel better not to know'" (Brainpickings.org). Heffernan says that the fear of conflict and change is what drives willful blindness. The fear of those things causes us to pretend that the situation is not a reality until, in most cases, it has already happened.

As I searched for a way to make some of these ideas more concrete, the bombing of Pearl Harbor immediately came to mind. Before Pearl Harbor, most of America was in denial about the situation across the Atlantic. Michael Morella put it best when touching on this subject, "Americans were listening to the radio, as all Americans did at that time. They were listening to local programming—news, farm reports, game shows, children's [programs], music contests, talent shows, things like that. In those days, the average American went to the movies twice a week. And the movies that they may have been seeing that night would have been Citizen Kane or The Maltese Falcon or Dumbo. But they weren't thinking about war. They really weren't thinking about war in the Pacific—that was the farthest thing from their minds. They were thinking about the war in Europe, but they weren't thinking about their involvement in the war in Europe." This scenario resembles the mindset of those living in Phoenix. It is easy for them to see that they are living in a desert, and not wonder where their water is from. They can hear about droughts in Africa, but ignore the same scenarios in their own backyards even as the grass turns brown and dies right before their eyes.

We know why humans choose to ignore future problems, pushing them off to the next generation. But why do places like New Orleans and Philadelphia differ? In America, we can safely assume that politics has something to with it. Although they turned in the Democrats favor this past election in 2020, Arizona voted Republican in the past three elections. New Orleans, which has a democratic mayor, normally elects Democrats in its local elections even with the state usually trending in favor of the Republicans. Philadelphia consistently votes blue in local, state, and federal elections. It is a well proven fact that both of the major political parties have their issues when it comes to political denial. These denials of facts come in two forms: People who deny the facts to themselves, and people who deny the facts to others. The latter situation is more insidious, because these "merchants of doubt" know the facts themselves very well. An example given in a 2019 article entitled "Climate explained: why some people still think climate change isn't real," written by David Hall, who is a senior researcher in politics at Auckland University of Technology (and who I must note seems to have Democratic views) is the ExxonMobile scenario.

“A comprehensive analysis of documentations from ExxonMobil found that, since 1977, the company has internally acknowledged climate change through the publications of its scientists, even while it publicly promoted doubt through paid advertorials. The fossil fuel industry has also invested heavily in conservative foundations and think tanks that promote contrarian scientists and improbable spins on the science.” If people are able to spread denial on purpose, then it is no wonder that people living in places like Phoenix can be in such denial by accident. In more blue-leaning places like Philadelphia, this type of denial is being met with a greater force of realism on the subject of climate change, and that is why the climate plan in that city is so robust.

The truth of the matter is, even if Philadelphia becomes the greenest city in the world and has 100% sustainability, and Phoenix becomes uninhabitable by 2050, like the Washington Post predicts, those two events by themselves have no real affect on climate change because neither one of these events is enough to “solve” the climate crisis. The good news is, the world is moving in the right direction, even if Arizona is not. The graph below from Energy.gov shows the reduction



in price of many of the technologies necessary to help transition away from climate-damaging energy sources. While the trend of clean energy sources getting cheaper is great in itself, its deeper meaning is much more exciting. These technologies are only getting cheaper due to the economies of scale surrounding them. In order for an LED to get cheaper, the entire global supply chain behind that light bulb needs to be invested in, grow, and scale. More people need to buy them, so that more people can afford to buy them. The graph above shows that, in at least the categories presented, there has been, and continues to be, demand and investment in the technologies necessary to carry out the plans set forth by cities everywhere.

The topics discussed in this paper are ever changing and evolving, and five years from now, in 2025, we could be in a very different place. But one things is certain: it is going to take more than documents and city initiatives to get out from under climate change. It is going to take the commitment and dedication of all the people on the planet. Humanity has never been tested to the degree that it will from now on. Survival is not the goal. Preservation of our way of life is the goal. Those with means can survive in a post climate change world because in order to do so you they will have access to an immense amount of resources that not everyone on this planet can afford. Pumping water uphill 300 miles cannot be reasonably scaled to every water insecure city. That is why those of us with means need to make the changes that those without cannot make themselves. It was once believed that we must make sacrifices in order to become sustainable, but this is not the case. We must make large sweeping changes, but with the right dedication and technology we will not be making sacrifices, we will be making advancements. Dr. Ernest Moniz, a nuclear physicist and Secretary of Energy under the Obama administration put of well: “Solving climate change is about the human spirit and our ability to tackle shared challenges together. It’s about ensuring energy security, expanding access to reliable and affordable energy, and spurring economic growth that creates jobs and protects the planet.”

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